National System Operator (Pvt) Ltd
Sri Lanka

POWER SYSTEM STRENGTHENING AND RENEWABLE ENERGY INTEGRATION PROJECT

(ADB LOAN NO: 4548-SRI)

PACKAGE 6

Procurement of Plant

Design, Supply, and Installation

Single-Stage: Two-Envelope Bidding Procedure

BIDDING DOCUMENT FOR

Integration of Renewable Energy Management System (REMS) and NSCC SCADA/EMS Modernization

VOLUME 4 of 8

Part II- REQUIREMENTS

Section 6 - Employer's Requirements: Part A-Scope of Works

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Issued on: March 30, 2026
Invitation for Bids No.: CEB/AGM(P)/TDE/IFB/2025/02/HLPC
OCB No.: CEB/AGM(P)/TDE/OCB/2025/02/HLPC
Employer: National System Operator (Pvt) Ltd
Country: Sri Lanka

Office of the Director (System Control)
National System Operator (Pvt) Ltd,
No. 80, Parliament Road,
Pelawatta, Battaramulla.
Sri Lanka.
Document – Revision 1

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Preface

This Bidding Document for Procurement of Plant – Design, Supply, and Installation, has been prepared by National Transmission Network Service Provider (Private) Limited and is based on the Standard Bidding Document for Procurement of Plant – Design, Supply, and Installation (SBD Plant) issued by the Asian Development Bank dated December 2021.

ADB's SBD Plant has the structure and the provisions of the Master Procurement Document entitled "Procurement of Plant – Design, Supply, and Installation", prepared by multilateral development banks and other public international financial institutions except where ADB-specific considerations have required a change.

In this Bidding Document, any reference to 'Ceylon Electricity Board' shall be deemed to refer to 'National System Operator (Pvt) Ltd.' Similarly, any reference to 'CEB' shall be deemed to refer to 'NSO'.

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Section 9 - Contract Forms (COF) ---------- **9-1**
This Section contains forms, which, once completed, will form part of the Contract. The forms for Performance Security and Advance Payment Security, when required, shall only be completed by the successful Bidder after contract award.

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1 SCOPE OF WORKS

1.1 INTRODUCTION

Sri Lanka's power sector is undergoing a fundamental transformation driven by increasing integration of Renewable Energy (RE), evolving regulatory requirements, regional interconnection ambitions, and the need for resilient, real-time grid operations. The Integration of a Renewable Energy Management System (REMS) and Modernization of the Supervisory Control and Data Acquisition (SCADA) / Energy Management System (EMS) platform at the National System Control Centre (NSCC) represents a cornerstone initiative to meet these challenges.

This project is aligned with the Government of Sri Lanka's vision for a clean energy transition and is structured under the Asian Development Bank (ADB) procurement framework for turnkey infrastructure projects. It aims to upgrade the technological backbone of national grid operations through a comprehensive set of system and infrastructure enhancements.

1.1.1. PROJECT CONTEXT AND DRIVERS

- Renewable Energy Growth: With a national target to reach 70% electricity generation from renewable sources by 2030, the grid must be equipped to handle the variability and decentralization of solar, wind, small hydro, and future Battery Energy Storage Systems (BESS).
- Aging Infrastructure: The current SCADA/EMS system at NSCC, based on "GE E-Terra" Platform, is limited in scalability, integration capability, and fails to support modern forecasting and Automatic Generation Control (AGC) features.
- Grid Complexity: With 80+ grid substations and 30+ major power stations connected using IEC 101 and 104 protocols, there is a pressing need for centralized, secure, and interoperable control.
- Regional Cooperation & Market Evolution: The future of Sri Lanka's power grid includes the possibility of cross-border tie-line operation, regional trading platforms (SAARC, BIMSTEC), and internal market mechanisms (e.g., ancillary services, demand response).

1.1.2. PROJECT OBJECTIVES

- Modernize the national SCADA/EMS system to enable full observability and controllability of the transmission network and generation stations.
- Integrate a REMS capable of forecasting, dispatch optimization, curtailment management, and compliance monitoring.
- Provide real-time, data-driven operational visibility and decision support tools for system operators.
- Enhance grid security through outage management and stability assessment tools.
- Enhance digital systems, network and data security through advanced cybersecurity enforcement.
- Future-proof the infrastructure for regional integration, tie-line control and market-based operations.

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1.1.3. GENARAL DESCRIPTION OF THE PROJECT

This document elaborates on the scope of work for each of the following systems and modules, which together comprise the proposed solution:

1. Operating Systems and Antivirus
2. SCADA/EMS System Hardware
   - Servers
   - Workstations
   - Office Personal Computers (Office PCs)
   - Control Centre Operations Display
3. SCADA Applications
4. EMS Applications
   - Modernization of the SCADA/EMS Master Station
   - Renewable Energy Management System (REMS)
   - Renewable Energy Forecasting System
   - Demand Forecasting System
   - Dispatch Scheduling and Automatic Generation Control (AGC)
   - Transparent Curtailment and Generation Management
   - Transmission Network Security Assessment
   - Battery Energy Storage System (BESS) Monitoring and Control
   - Real-time Operational Requirements Assessment
   - Power Plant and Network Outage Management
   - Implementation of Cybersecurity Enforcement
   - Provisions for Market Integration and Cross-Border Interconnection
5. Supplementary Applications
   - Switching Schedules
   - Job Management
   - Under-Frequency Load Shedding (UFLS)
   - Load Shedding Schedules
   - Historical Data
   - Report Generation
   - Operator Training Simulator (OTS)
6. Remote Station Adaptation Work
   - Exiting Grid Substations
   - Exiting Power Plants

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- Existing Grid Support Services

7. Migration from Existing System to New Modernized SCADA/EMS/REMS

1.1.4. CONTRACTUAL FRAMEWORK

This project is executed as a Turnkey Design-Supply-Install-Test-Commission contract. The Bidder will be fully responsible for system design, site surveys, factory and site testing, documentation, training, and full system commissioning with measurable performance guarantees.

1.1.5. STRATEGIC OUTCOMES

Upon successful completion, the modernized NSCC will:
- Ensure seamless integration and operation of high-penetration renewables.
- Provide fast, reliable response to grid disturbances.
- Improve transparency and accountability in system operation and curtailment decisions.
- Enable participation in regional energy markets and future grid services (e.g., frequency response, load shifting).
- Strengthen resilience against cyber threats and enhance business continuity planning.

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1.1.6. SCOPE OF WORKS RELATED TO REMOTE END GRID SUBSTATIONS (GS), POWER STATIONS (PS) AND GRID SUPPORT SERVICES

Design services to integrate all GSs, PSs and Grid support services to NSCC including,

- Protocol adaptation
- Signal data definition
- Any other design aspects

Commissioning to integrate all GSs, PSs and Grid support services to NSCC including,

- Alarms simulation
- Point to point testing for indications and controls
- Measurement verification

1.1.7. TERMINAL POINTS

The following shall be the contract terminal points: -

1. Within NSCC

(a) IEC 60870 – 5 – 101 and IEC 60870 – 5 – 104 protocols related to GSs and PSs are terminated at the communication panel at NSCC. This contract includes integrating these SCADA data to proposed SCADA/ EMS.

(b) Inter-control Centre Communication Protocols (ICCP) related to Substation E and Substation F in Colombo City and Distribution Control Centres (DCC) are terminated at the communication panel at NSCC. This contract includes integrating these SCADA data to proposed SCADA/ EMS.

(c) File Transfer Protocol (FTP)/ ICCP data related to CEB corporate network (CEB Assist network) are terminated at the CEB corporate LAN at NSCC. This contract includes integrating these SCADA data to proposed SCADA/ EMS.

(d) This contract includes the modifications of existing UPS LVAC system to obtain necessary power requirements.

2. Remote GSs, PSs and Grid Support Services

(a) IEC 60870 – 5 – 101 and IEC 60870 – 5 – 104 protocols related to GSs, PSs and Grid support services are terminated at the Gateway/ RTU at the remote GSs, PSs and Grid support services. This contract includes integrating these SCADA data to proposed SCADA/ EMS at NSCC.

(b) Inter-control Centre Communication Protocols (ICCP) related to Substation E and Substation F in Colombo City and Distribution Control Centres (DCC) are terminated at the ICCP server. This contract includes integrating these SCADA data to proposed SCADA/ EMS at NSCC.

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1.1.8. SITE ENVIRONMENTAL CONDITIONS

Colombo

Altitude of site above sea level not exceeding	m	1000
Maximum ambient air temperature	^{0}C	36.1
Minimum ambient air temperature	^{0}C	18.2
Average ambient air temperature over 24 hours	^{0}C	30
Average yearly temperature	^{0}C	27.6
Relative humidity (24 hours)	%	81
Average annual rainfall	mm	2424
Maximum recorded rainfall for 24 hours	mm	494
Thunderstorms	Frequency of thunderstorms is high	
Average isokeraunic level	110 days per annum	
Atmospheric corrosive category	C4 (ISO 12944-2)	
Pollution level for bushings and insulators	43.3 mm/kV (USCD)	

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1.1.9. ELECTRICITY, WATER, GAS AND OTHER SERVICES

The Bidder shall at his expense, provide all electricity, water, gas and other services necessary to execute and complete the Works on site. Prevailing tariff and service connection procedure shall be applicable.

1.1.10. ADHERENCE TO THE ENVIRONMENTAL MITIGATION MEASURES

Bidders are requested to comply with the requirements stated in the Chapter 1, Clause 1.8 of Section 6 Part B -Technical Specifications and also to the environmental safeguard measures described in the following documents attached in the section 6 Employers requirement Part D- Supplementary Information.

The said documents prepared in accordance with the governing acts and/or guidelines, which are particularly relevant to this project may be inspected and a copy obtained free of charge from the office of the Director (System Control) at the following address.

Address:

Office of the Director (System Control)
National System Operator (Pvt) Ltd,
No. 80,
Parliament Road,
Pelawatta,
Battaramulla 10120
Sri Lanka

Telephone : (+94) 11 278 7950
Facsimile number : (+94) 11 278 7948
Electronic mail address : dgmsyc@nso.lk

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1.2 PLANT & EQUIPMENT INCLUDING MANDATORY SPARE PARTS SUPPLIED OUTSIDE THE EMPLOYER'S COUNTRY

1.2.1 SCADA/EMS MASTER STATION SYSTEM SOFTWARE

Bidder shall provide the latest versions of SCADA/EMS/REMS application software, Operating Systems (OS) available at the time of implementation.

1.2.1.1 HYPERVISOR PLATFORMS FOR SERVER CLUSTERS

Each virtualization host shall operate on an enterprise-grade Type-1 hypervisor supporting real-time, multi-processor, and multitasking operations. The platform shall ensure high availability, resource isolation, and deterministic performance for SCADA/EMS/REMS applications, with support for secure boot, role-based access, and live migration in compliance with IEC 62443 cybersecurity requirements.

1.2.1.2 OPERATING SYSTEMS FOR VIRTUAL MACHINES/ WORKSTATIONS

Each virtual machine unit (servers, workstations, FEPs, gateways) shall run a real-time, multi-user, multitasking OS capable of executing SCADA/EMS/REMS applications reliably.

1.2.1.3 ANTIVIRUS SOFTWARE

All Master Station computer systems, including servers, all virtual machines, operator workstations, and auxiliary consoles, shall be supplied with enterprise-grade antivirus and Endpoint Detection and Response (EDR) software.

Software license validity period shall be minimum 3 years after defect liability period.

1.2.1.4 MISCELLANEOUS WORKS

Any work other than listed above.

1.2.1.5 SPARE PARTS

All necessary Spare parts.

1.2.1.6 TOOLS AND INSTRUMENTS

As per the price schedule.

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1.2.2 SCADA/EMS MASTER STATION HARDWARE

1.2.2.1 SERVERS (VIRTUALIZED ENVIRONMENT)

All servers deployed for the Master Station shall be of proven enterprise-class design, suitable for continuous real-time operation where reliability and low latency are paramount. The servers shall be built to support virtualisation.

1.2.2.1.1 Enterprise Server Cluster (Web/ Patch/ Historian)

02 Nos - rack-mounted servers with load balancing and high-availability configurations as per Chapter 18 General Technical Specification.

1.2.2.1.2 Application/ SCADA Server Cluster

04 Nos - rack-mounted servers with two clusters designed with load balancing and high-availability configurations as per Chapter 18 General Technical Specification.

1.2.2.1.3 Security Operations Centre (SOC) Server Cluster

02 Nos - rack-mounted servers with load balancing and high-availability configurations as per Chapter 18 General Technical Specification.

1.2.2.1.4 Front End Processor (FEP) Server Cluster

02 Nos - rack-mounted servers with Load balancing and high-availability configurations as per Chapter 18 General Technical Specification.

1.2.2.1.5 OTS Server Cluster

02 Nos - rack-mounted servers with Load balancing and high-availability configurations as per Chapter 18 General Technical Specification.

1.2.2.1.6 Other

Any other servers required for completion of the SCADA/EMS/REMS system. The requirement of the additional servers shall be provided with the technical Bid.

1.2.2.2 WORKSTATIONS

All workstations shall replicate the full SCADA/EMS user interface, each equipped with ergonomic three displays, alphanumeric keyboard, and pointing device.

1.2.2.2.1 Operator Workstations

05 Nos – High performance workstations with three (03) multi monitor displays as per Chapter 18 General Technical Specification.

1.2.2.2.2 Maintenance Workstation

01 No – High performance workstations with three (03) multi monitor displays as per Chapter 18 General Technical Specification.

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1.2.2.2.3 Power System Analysis Workstations

06 Nos – High performance workstations with three (03) multi monitor displays as per Chapter 18 General Technical Specification.

1.2.2.2.4 Office PCs

10 Nos – High performance PCs as per Chapter 18 General Technical Specification.

1.2.2.3 CONTROL CENTRE OPERATIONAL DISPLAY AND INFRASTRUCTURE

1.2.2.3.1 Control Centre Operational Display

1 Lot – As per the Chapter 18 of General Technical Specification

1.2.2.3.2 System Displays

1 Lot – As per the Chapter 18 of General Technical Specification

1.2.2.4 LOCAL AREA NETWORK (LAN) - SWITCHES, ROUTERS, & CABLING

Local Area Network shall be provided as per Chapter 18 General Technical Specification and typical architecture drawing provided in Volume 06 of bidding document.

All Ethernet Switches shall have redundant AC power supplies. Each switch shall have at least five copper RJ45 spare ports. All the un-utilized ethernet ports shall be kept disabled, as a measure to increase Cybersecurity.

1.2.2.5 FIREWALL

Firewall shall be provided as per Chapter 18 General Technical Specification and typical architecture drawing provided in Volume 06 of bidding document.

1.2.2.6 STORAGE AREA NETWORK (SAN)

Storage Area Network (SAN) shall be provided as per Chapter 18 General Technical Specification and typical architecture drawing provided in Volume 06 of bidding document.

1.2.2.7 GPS TIME SERVER

01 Lot. - GPS Time Server as per Chapter 18 General Technical Specification

1.2.2.8 PRINTERS AND SCANNERS

02 Nos – Colour printers as per Chapter 18 General Technical Specification.

02 Nos – Monochrome printers as per Chapter 18 General Technical Specification.

02 Nos – Scanners as per Chapter 18 General Technical Specification.

1.2.2.9 MISCELLANEOUS WORKS

Any work other than listed above.

1.2.2.10 SPARE PARTS

All necessary Spare parts.

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1.2.2.11 TOOLS AND INSTRUMENTS

As per the price schedule.

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1.2.3 SCADA/EMS MASTER STATION APPLICATIONS

1.2.3.1 SCADA APPLICATIONS

The modernization of the existing SCADA/EMS Master Station at the National System Control Centre (NSCC) aims to enhance system reliability, operational capability, and flexibility in managing the national transmission and generation system. The current system, based on the GE e-Terra Platform, connects to approximately 80 grid substations and 30 major power stations using IEC 60870-5-104 and IEC 60870-5-101 protocols, with an average of 500 data points per substation. The scope includes complete replacement of the master station software, ensuring scalability and compliance with modern operational and cybersecurity standards.

1.2.3.1.1 Functional Requirements

SCADA Core Functionality

- Acquisition, processing, and visualization of real-time data (status, analog, alarms, commands) from substations and power plants in accordance with Annex A, B, C, D, E and F.
- Support for IEC 60870-5-104, IEC 60870-5-101, ICCP/TASE.2, IEC 61850 and HTTP/S (web-based API) protocols
- Minimum capacity of 150,000 real-time points with scalability for future expansion
- Advanced HMI/GUI for operators with zoomable geographic, single-line, and tabular displays
- Time synchronization of all services related to SCADA/EMS system shall use GPS/NTP and support for event timestamping (SOE) with 1 ms accuracy
- AI-assisted analytics for wide-area situational awareness

System Redundancy and High Availability

- Minimum dual redundant SCADA/EMS application services (Hot-Standby or Active-Active) with automatic failover and data replication
- Redundant communication front-end processor (FEP) applications
- Real-time data replication and failover testing support
- Physical Separation of Redundant Devices

All redundant devices forming part of the Master Station infrastructure, including but not limited to firewalls, servers, storage systems, network switches, routers, time servers, and security appliances, shall be installed in physically separate locations within the NSCC premises. These locations shall be independent in terms of physical environment, power supply, cooling systems, fire detection and suppression, access control, and environmental monitoring, in order to eliminate the risk of common-mode

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physical failures.

Historical and Reporting Systems

- Integrated historian system for long-term storage of real-time, alarm, and event data using SAN
- Support for trend analysis, system logs, record retrieval using SQL and custom report generation
- Export of data to regulatory and planning tools in CSV/XML formats

Cybersecurity and Access Control

- Implementation of cybersecurity measures per IEC 62443 and NIST SP 800-82
- Role-Based Access Control (RBAC), audit trails, and Multi-Factor Authentication (MFA) support for operator logins
- Patch management and anti-malware protection for all SCADA/EMS components
- All cyber security measures including authentication and authorization shall be handled by SOC

1.2.3.1.2 Implementation Requirements

- Site survey and system audit of existing GE e-Terra based SCADA/EMS configuration
- Detailed migration plan with minimal downtime and staged switchover
- Data point mapping, communication testing, and site integration for all substations, power plants and distribution control centres in accordance with Annex A, B, C, D, E and F
- Factory Acceptance Testing (FAT), Site Acceptance Testing (SAT), and System Integration Testing (SIT)
- Documentation covering system architecture, configuration, backup/recovery, and maintenance

1.2.3.1.3 Future-Proofing and Expansion

- Scalability to add new substations, new power stations, new DCCs and new gird support services in accordance with Annex D, E and F
- Modular software architecture to support new applications (e.g., REMS, market management systems)
- Remote access capabilities for disaster recovery centre

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1.2.3.2 ENERGY MANAGEMENT SYSTEM

01 Lot. – Comprehensive Energy Management System as per chapter 18 of General Technical Specification

1.2.3.3 SUPPLEMENTARY APPLICATIONS

01 Lot. – Supplementary applications as per chapter 18 of General Technical Specification

1.2.3.4 MISCELLANEOUS WORKS

Any work other than listed above.

1.2.3.5 SPARE PARTS

All necessary Spare parts.

1.2.3.6 TOOLS AND INSTRUMENTS

As per the price schedule.

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1.2.4 OTHER INFRASTRUCTURE AND AUGMENTATION WORKS

1.2.4.1 CONTROL ROOM FURNITURE

1.2.4.1.1 Operator Desks and Chairs

01 Lot. – as per Volume 6: Plans and Drawings and General Technical Specification

1.2.4.1.2 Equipment Stands (to be specified by the bidder)

01 Lot. – to be specified by the bidder

1.2.4.2 All additional Cubicles & Ancillary Equipment

01 Lot. – All additional Cubicles & Ancillary Equipment required for the complete installation

1.2.4.3 Modifications to LVAC Panel and Distribution Boards

01 Lot. – Modifications to exiting distribution boards to complete augmentation works

1.2.4.4 Cables

01 Lot. – All required cables to complete augmentation works

1.2.4.5 MISCELLANEOUS WORKS

Any work other than listed above.

1.2.4.6 SPARE PARTS

All necessary Spare parts.

1.2.4.7 TOOLS AND INSTRUMENTS

As per the price schedule.

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1.3 CIVIL WORKS, INSTALLATION, OTHER SERVICES AND TRAINING

1.3.1 CIVIL WORKS

1.3.1.1 WORKS WITHIN NSCC

1.3.1.1.1 Mobilization & demobilization of machinery during the implementation

01 Lot. – as per General Technical Specification

1.3.1.1.2 Temporary sheds for stores and clearing after completion

01 Lot. – as per General Technical Specification

1.3.1.1.3 Setting out works

01 Lot. – as per General Technical Specification

1.3.1.1.4 Modifications to existing raised floor, cable trenches and trays

01 Lot. – Any modifications to raised floor, cable trenches and trays to complete works

1.3.1.1.5 Raised floor for Control room

01 Lot. – as per General Technical Specification. A wooden finished raised floor required

1.3.1.1.6 Modifications to Floor, wall & Ceiling finishes & fittings

01 Lot. – Any modifications to Floor, wall & Ceiling finishes & fittings to complete works

1.3.1.1.7 Electrical works

01 Lot. – Any modifications of electrical works

1.3.1.1.8 Mechanical works

01 Lot. – Any modifications of mechanical works

1.3.1.1.9 Miscellaneous Works

Any additional items Bidder may consider necessary to complete the installation

1.3.1.2 REMOTE ENDS

1.3.1.2.1 Any Civil Works Related to Remote Ends

Any items Bidder may consider necessary to complete the project

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1.3.2 INSTALLATION, TESTING AND COMMISSIONING

1.3.2.1 WORKS WITHIN NSCC

1.3.2.1.1 Installation, Testing and commissioning of SCADA/EMS/REMS System

1 lot - All the installation, testing and commissioning works for the completion of the project.

1.3.2.1.2 Migration from Existing to New Modernized SCADA/EMS/REMS

Bidder shall provide detailed migration plan for the migration of exiting system to new modernized SCADA/EMS/REMS system. All items used for migration shall be handed over to NSO after successful migration.

1.3.2.1.2.1 Data engineering work

Data engineering work for transferring existing SCADA/EMS data and diagrams to new system

1.3.2.1.2.2 Transition from existing to new systems

Work associated with facilitating transition from existing to new systems including temporary data link between new and existing SCADA Master Stations.

1.3.2.1.2.3 Additional Hardware, Tools and Test Apparatus for Migration

Bidder shall include any additional tools or test apparatus that Bidder may consider necessary to ensure the correct operation and migration of the SCADA and Power Supply System.

Additional two Nos of industrial grade LED displays (Minimum 65 inch) are required during the replacement of control centre operational display for the continuation of system operations.

1.3.2.1.3 Miscellaneous Works

The Bidder shall include any additional migration items may consider necessary for the provision of a full and complete migration of facilities by maintaining daily operational work.

1.3.2.2 INTEGRATION OF REMOTE ENDS

1.3.2.2.1 Migration of Existing Grid Substations and DCCs

Migration of Existing Grid Substations and DCCs as per Annex A. The Bidder shall provide all necessary applications and tools required to facilitate data replication/ mirroring between the existing system and the new SCADA/EMS system.

1.3.2.2.2 Migration of Existing Major Power Stations and NCRE Power Stations

Migration of Existing Major Power Stations and NCRE Power Stations as per Annex B. The Bidder shall provide all necessary applications and tools required to facilitate data replication/ mirroring between the existing system and the new SCADA/EMS system.

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1.3.2.2.3 Migration of Existing Grid Support Services

Migration of Existing Grid Support Services as per Annex C. The Bidder shall provide all necessary applications and tools required to facilitate data replication between the existing system and the new SCADA/EMS system.

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1.3.3 TRAINING

Local Training: All local training programmes shall be conducted on the proposed system to ensure effective hands-on learning and operational familiarity.

Foreign Training: Foreign training programmes shall be undertaken after the implementation of the contract, enabling participants to gain maximum benefit from training based on an operational environment.

1.3.3.1 System overview (in Sri Lanka)

01-day session in Sri Lanka for 20 persons

1.3.3.2 System Operation and Application Training Courses (in Sri Lanka)

03 Nos of sessions in Sri Lanka each for 05 days and 05 persons

1.3.3.3 Foreign Trainings

1.3.3.3.1 Database & Display Maintenance Training Courses

10 days and 08 persons

These training programmes shall be undertaken prior to the commissioning works.

1.3.3.3.2 EMS/REMS Application Training Courses

10 days and 10 persons

These training programmes shall be undertaken prior to the commissioning works.

1.3.3.3.3 System Administration and Maintenance Training Courses

10 days for 06 persons

These training programmes shall be undertaken during the site implementation of new SCADA/EMS/REMS system with the consent of the Employer.

1.3.3.4 Any Additional Training

The Bidder shall include any additional trainings that may consider necessary to ensure the correct operation and maintenance of the SCADA/EMS/REMS system

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1.3.4 OTHER SERVICES

1.3.4.1 TRANSPORT FOR EMPLOYER'S PERSONNEL

1.3.4.1.1 Four Wheel Drive (Off Road)

03 Nos of Four-Wheel Drive (Off Road)

1.3.4.1.2 Running cost

Running costs for 36,000 km / year for each vehicle in clause 1.3.4.1.1

1.3.4.2 PROJECT OFFICE ACCOMMODATION FOR EMPLOYER'S PERSONNEL

1.3.4.2.1 Employer's office accommodation

Minimum 1500 square feet accommodation nearby NSCC.

Shall include a meeting room for minimum 15 persons

1.3.4.3 ENVIRONMENTAL MITIGATION MEASURES

Environmental mitigation measures according to clause 1.1.9 of part a - scope of works, section 6 - employer's requirements.

1.3.4.4 ENVIRONMENTAL MANAGEMENT PLAN

1.3.4.5 MAINTENANCE AGREEMENT

The Bidder shall carry out the mandatory maintenance activities for Master Station hardware, system software and all application software during the 2-year defect liability period. Competent personnel shall be deployed for such maintenance activities. Knowledge transfer and capacity building of Employer's staff shall be ensured in order to carry out the maintenance activities of the Master Station hardware, system software and all application software after the 2-year defect liability period. These routine maintenance training programs and schedules are subjected to the prior approval of the Employer.

1.3.4.5.1 Maintenance Agreement for subsequent three years

Maintenance Agreement for Master Station hardware, system software and all application software for subsequent three years after 2-year defect liability period.

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ANNEX A: EXISTING GRID SUBSTATIONS AND DCCS

No	EXISING GRID SUBSTATION NAME
1	AMBALANGODA GS
2	AMPARA GS
3	ANIYAKANDA GS
4	ANURADHAPURA GS
5	ATHURUGIRIYA GS
6	BADULLA GS
7	BALANGODA GS
8	BELIATTE GS
9	BIYAGAMA GS
10	BOLAWATTE GS
11	CHILLAW (MADAMPE) GS
12	CHUNNAKAM GS
13	DEHIWELA GS
14	DENIYAYA GS
15	EMBILIPITIYA GS
16	HABARANA GS
17	HAMBANTOTA GS
18	HORANA GS
19	KALUTARA GS
20	KAPPALTURAI GS
21	KATUNAYAKE GS
22	KEGALLE GS
23	KELANITISSA 220/132/33 GS
24	KELANIYA GS
25	KERAWALAPITIYA GS
26	KESBEWA GS
27	KILINOCHCHI GS
28	KIRIBATHKUMBURA GS
29	KIRINDIWELA GS
30	KOLONNAWA GIS
31	KOSGAMA GS
32	KOTUGODA 220/132/33 GS
33	KURUNEGALA GS
34	MAHIYANGANA GS
35	MAHO GS
36	MALIBODA GS
37	MANNAR GS
38	MATARA GS
39	MATUGAMA GS
40	MONARAGALA GS
41	NADUKUDA GS
42	NAULA GS
43	NAWALPITIYA GS
44	NEW ANURADHAPURA
45	NEW CHILLAW GS

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No	EXISING GRID SUBSTATION NAME
46	NEW GALLE GS
47	NEW HABARANA GS
48	NEW KERAWALAPITIYA
49	NEW POLPITIYA GS
50	NSCC
51	NUWARA ELIYA GS
52	ORUWELA GS
53	PADUKKA GS
54	PALLEKELLE GS
55	PANADURA GS
56	PANNALA GS
57	PANNIPITIYA 220/132/33 kV GS
58	POLONNARUWA GS
59	PUTTLAM GS
60	RAGALA GS
61	RATMALANA GS
62	RATNAPURA GS
63	SAPUGASKANDE GS
64	SEETHAWAKA GS
65	SRI JAYAWARDANAPURA GS
66	SUB A (HAVELOCK TOWN)
67	SUB B (FORT)
68	SUB C (BLOEMENDHALL)
69	SUB E (KOLLUPITIYA)
70	SUB F (FORT)
71	SUB I (MARADANA)
72	SUB L
73	SUB M
74	SUB N
75	SUB R1
76	THULHIRIYA GS
77	TRINCOMALEE GS
78	UKUWELA GS
79	VALACHCHENAI GS
80	VAVUNATIVU GS
81	VAVUNIYA GS
82	VEYANGODA 220 kV GS
83	WAGAWATTA GS
84	WEWELWATTE GS
85	WIMALASURENDRA GS

No	EXISING DCC
86	Colombo City
87	Dehiwala

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ANNEX B: EXISTING MAJOR POWER STATIONS AND NCRE POWER STATIONS

No	EXISING MAJOR POWER SUBSTATION NAME
1	BARGE PS
2	BOWATENNA PS
3	BROADLANDS PS
4	CANYON PS
5	DAKUNU JANANI PS
6	INGINIYAGALA PS
7	KCCP PS
8	KCCP2 PS
9	KERAWALAPITIYA WASTE TO ENERGY IPP
10	KOTMALE PS
11	KPS PS
12	KUKULE PS
13	LAKVIJAYA PS
14	LAXAPNA OLD PS
15	MATUGAMA DPP PS
16	MORAGAHAKANDA IPP
17	MORAGOLLA PS
18	NEW LAXAPANA PS
19	NILAMBE PS
20	POLPITIYA PS
21	RANDENIGALA PS
22	RANTEMBE PS
23	SAMANALAWEWA PS
24	SAPUGASKANDA A PS
25	SAPUGASKANDA B PS
26	SOBADHANAVI IPP
27	THAMBAPAVANI PS
28	UDAWALAWA PS
29	UKUWELA PS
30	UMAOYA PS
31	UPPER KOTMALE PS
32	UTHURU JANANI PS
33	VICTORIA PS
34	WEST COAST POWER IPP
35	WIMALASURENDRA PS

No	EXISING NCRE POWER STATION TECHNOLOGY	No OF PLANTS
36	Mini Hydro	219
37	Solar Power	99
38	Wind	22
39	Dendro and Biomass	14
40	Municipal Solid Waste	1

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ANNEX C: EXISTING GRID SUPPORT SERVICES

No	Existing Grid Support Services	No. of Plants
1	Variable Shunt Reactor (VSR)	3
2	Static Synchronous Compensator (STATCOM)	1
3	Breaker Switch Capacitor (BSC) GS	25
4	Frequency Controlling Units	12

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ANNEX D: FUTURE GRID SUBSTATIONS AND DCCS

No	FUTURE GRID SUBSTATION NAME
1	Wariyapola 220/132kV
2	Wariyapola 132/33kV
3	Ekala 132/33kV
4	Kandy City 132/11kV
5	Kalawana 132/33kV
6	Negombo 132/33kV
7	Mirigama 220/33kV
8	Tissamaharama 132/33kV
9	Baddegama a 132/33kV
10	Homagama 132/33kV
11	Peliyagoda 132/33kV
12	Colombo G 220/132kV
13	Colombo K 132/33kV
14	Colombo P 132/33kV
15	Colombo Q 132/33kV
16	Nothern Collector 220kV
17	Port City R2 220/33kV
18	Chemmani 220/33kV
19	Keeriyankalliya 132/33kV

No	FUTURE DCC
20	Distribution Division 1
21	Distribution Division 2
22	Distribution Division 3
23	Distribution Division 4
24	LECO

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ANNEX E: FUTURE POWER STATIONS

No	FUTURE POWER SUBSTATION TECHNOLOGY	No OF PLANTS
1	Solar	150
2	Wind	50
3	MHP	50
4	Waste Energy	10
5	Bio Mass	20

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ANNEX F: FUTURE GRID SUPPORT SERVICES

No	Future Grid Support Services	No. of Plants Or GS
1	Static Synchronous Compensator (STATCOM)	4
2	Variable Shunt Reactor (VSR)	4
3	Battery Energy Storage System (BESS)	50
4	Breaker Switch Capacitor (BSC)	10
5	Synchronous Condenser (SynCon)	4
6	HVDC Tie Line	1
7	Frequency Controlling Units	4
8	Pump Storage	2

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